



Koneru Lakshmaiah Education Foundation

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Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

WEEK – 2

The Librarian's Dashboard Scenario The library is now digital! The head librarian needs a report to see which members are active and which books are in the collection. You also need to handle a "Book Donation" process where a new book is added and assigned to a category at the exact same time—if one part fails, the whole process must stop to keep the catalog accurate.

```
Microsoft Windows [Version 10.0.26200.8875]
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C:\Users\pothu>cd C:\Program Files\MySQL\MySQL Server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE bookmanagement_db;
Query OK, 1 row affected (0.03 sec)

mysql> USE bookmanagement_db;
Database changed
mysql> CREATE TABLE Books (
  -> book_id INT PRIMARY KEY,
  -> title VARCHAR(100) NOT NULL,
  -> isbn VARCHAR(20) UNIQUE,
  -> published_year INT CHECK(published_year<2027)
  -> );
Query OK, 0 rows affected (0.31 sec)
```

```
mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES
-> (1, 'THE GREAT GATSBY', '978074273565', 1925),
-> (2, 'TO KILL A MOCKINGBIRD', '9780061120084', 1960),
-> (3, '1984', '9780451524935', 1949);
```

Query OK, 3 rows affected (0.07 sec)
Records: 3 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM Books;
```

book_id	title	isbn	published_year
1	THE GREAT GATSBY	978074273565	1925
2	TO KILL A MOCKINGBIRD	9780061120084	1960
3	1984	9780451524935	1949

3 rows in set (0.01 sec)

```
mysql> CREATE TABLE Members (
-> member_id INT PRIMARY KEY,
-> full_name VARCHAR(100),
-> email VARCHAR(100) UNIQUE
-> );
```

Query OK, 0 rows affected (0.10 sec)

```
mysql> INSERT INTO Members (member_id, full_name, email) VALUES
-> (101, 'John Smith', 'johnsmith@email.com'),
-> (102, 'Emma Wilson', 'emma.willson@email.com'),
-> (103, 'Micheal Brown', 'micheal.brown@email.com');
```

Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM Members;
```

member_id	full_name	email
101	John Smith	johnsmith@email.com
102	Emma Wilson	emma.willson@email.com
103	Micheal Brown	micheal.brown@email.com

3 rows in set (0.00 sec)

```
mysql> CREATE TABLE Loans (
-> loan_id INT PRIMARY KEY,
-> member_id INT,
-> book_id INT,
-> loan_date DATE,
-> FOREIGN KEY (member_id) REFERENCES Members(member_id),
-> FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
```

Query OK, 0 rows affected (0.11 sec)

```
mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
-> (1, 101, 1, '2025-01-05'), (2, 101, 2, '2025-01-08'),
-> (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
-> (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
-> (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-02-07'),
-> (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
Query OK, 10 rows affected, 1 warning (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 1
```

```
mysql> SELECT * FROM Loans;
+-----+-----+-----+-----+
| loan_id | member_id | book_id | loan_date |
+-----+-----+-----+-----+
| 1 | 101 | 1 | 2025-01-05 |
| 2 | 101 | 2 | 2025-01-08 |
| 3 | 103 | 3 | 2025-01-10 |
| 4 | 101 | 2 | 2025-02-01 |
| 5 | 102 | 1 | 2025-02-05 |
| 6 | 103 | 2 | 2025-02-12 |
| 7 | 101 | 3 | 2025-03-01 |
| 8 | 102 | 3 | 2025-02-07 |
| 9 | 103 | 1 | 2025-03-15 |
| 10 | 101 | 1 | 2025-04-01 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> SELECT m.full_name AS Member_Name, b.title AS Book_Title
-> FROM Loans l
-> INNER JOIN Members m ON l.member_id = m.member_id
-> INNER JOIN Books b ON l.book_id = b.book_id;
+-----+-----+
| Member_Name | Book_Title |
+-----+-----+
| John Smith | THE GREAT GATSBY |
| John Smith | TO KILL A MOCKINGBIRD |
| John Smith | TO KILL A MOCKINGBIRD |
| John Smith | 1984 |
| John Smith | THE GREAT GATSBY |
| Emma Wilson | THE GREAT GATSBY |
| Emma Wilson | 1984 |
| Micheal Brown | 1984 |
| Micheal Brown | TO KILL A MOCKINGBIRD |
| Micheal Brown | THE GREAT GATSBY |
+-----+-----+
10 rows in set (0.01 sec)
```

```
mysql> SELECT published_year, COUNT(book_id) AS Total_Books
-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;
```

published_year	Total_Books
1925	1
1949	1
1960	1

```
3 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Donation_History (
-> donation_id INT PRIMARY KEY,
-> book_id INT,
-> donor_name VARCHAR(100),
-> donation_date DATE,
-> FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
```

```
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> START TRANSACTION;
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
-> VALUES (1, 4, 'RAJ KUMAR', CURDATE());
```

```
Query OK, 1 row affected (0.00 sec)
```

```
mysql> COMMIT;
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE INDEX idx_books_isbn ON Books(isbn);
```

```
Query OK, 0 rows affected (0.04 sec)
```

```
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Books WHERE isbn = '9780451524935';
```